

ABSTRACT

On 67 pages, 16 figures, 4 tables, 2 appendices

Keywords: high-throughput sequencing, degenerate PCR primers, ensemble thermodynamics, soil microbiota, rhizobia

The subject of the graduate qualification work is «Development of Software for Soil Microbiota Analysis Using High-Throughput Sequencing Systems».

The subject of the graduate qualification work is the development of software for soil microbiota analysis using high-throughput sequencing systems. The goal is to create high-performance software in Julia for designing degenerate PCR primers considering ensemble thermodynamics and its testing on rhizobia populations. The object of the research is the processes of bioinformatic analysis of sequencing data. The subject is the algorithms for calculating statistical distributions of thermodynamic parameters for pools of degenerate oligonucleotides. The methodology is based on the nearest-neighbor thermodynamic model, dynamic programming algorithms for secondary structure prediction, and probabilistic data denoising. As a result, the SeqFold.jl computational core was ported and optimized, and the DePPA.jl package was designed and implemented. Primer sets for six rhizobia genes were designed and laboratory validated. 36 amplicon libraries were prepared and successfully sequenced. Bioinformatic analysis confirmed the biological hypothesis: free-living soil populations form an extensive reservoir of unique alleles of symbiotic genes, which passes through a strict ecological filter of the host plant during symbiosis formation. The application area is bioinformatics, molecular ecology, and microbiology. The developed tools allow minimizing PCR biases in the targeted analysis of complex microbial communities.